Vickers Hardness Testing Lab

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ME 241 Materials Laboratory

Group B

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Abstract

In this lab we measured the Vickers hardness of five materials. Materials used included 6061-T6 aluminum, 1018 cold-rolled steel, 360 half-hard brass, 110 copper and a zirconia (ZrO_2) ceramic. The diagonals of each indentation were measured through the eyepiece of a Phase II Model 900-398 macro Vickers tester, which indented each specimen with a square-based diamond pyramid at a test force of 98.07 N (10 kgf) for 15 s. Each material was indented three times and six diagonal readings were taken. The average deviation of the six readings was then carried into hardness as an estimate of uncertainty. Results were 107 ± 2 HV10 for aluminum, 250 ± 10 for steel, 141 ± 5 for brass, 108 ± 3 for copper and 1600 ± 200 for zirconia. Zirconia was the hardest material and had the largest scatter.

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Introduction

Hardness is a measure of how resistant a material is to plastic deformation, such as denting or scratching. In an indentation test, a hardness testing machine pushes an indenter into the surface under a controlled load and the size of the impression or the depth of the impression is translated to a hardness number. A softer material makes a larger impression and gets a lower hardness number. Hardness tests are extremely inexpensive and leave only a slight dent, so they are frequently used as an inexpensive way to compare materials. For many metals, hardness also is dependent on tensile strength, as both are related to how much a metal resists plastic deformation.

Plastic deformation in a crystalline metal is dependent on the motion of dislocations in the structure of that material. The hardness of a metal depends on the amount of stress necessary to move these dislocations. When the motion of the dislocations are impeded, the metal becomes harder and stronger. Grain boundaries, solute atoms that distort the surrounding lattice, additional dislocations generated by cold working, and precipitate particles are all features that can impede that motion. Ceramic boundaries, however, resist slip for a different reason. In a crystal with ionic bonding, slip in some directions would bring ions of the same charge close together, and the electrostatic repulsion between them impedes the slip. This is why many ceramics are very hard and brittle.

The four metals in this lab employ different combinations of these mechanisms. The 110 alloy is electrolytic copper. Pure copper is soft and ductile. Cold work is one of the most important methods for hardening copper. The lab handout says this bar is 110 copper (however does not say what its temper is). The 360 brass is a free-cutting copper alloy containing copper, lead and zinc. Zinc atoms substitute for the copper atoms in the solid solution and help to

strengthen the material. The half-hard temper also introduces cold work. The hardness of the steel 1018 also results from cold working. The T6 condition of aluminum 6061 uses precipitation hardening.

The Vickers test uses a square-based diamond pyramid whose opposite faces meet at 136°.[1] The tester presses the indenter into the specimen under a test force F, holds the force, and removes it. The operator then measures both diagonals of the square impression under a microscope. The mean diagonal of each indent is:

$$d = (d1 + d2)/2 \tag{1}$$

The Vickers hardness number is the test force divided by the sloped contact area of the impression. A square with diagonal d has side a = d/sqrt(2) and plan area d^2/2. Opposite faces meet at 136°, so each face makes an angle of (180 - 136)/2 = 22 degrees with the specimen surface. The sloped area of the four faces equals the plan area divided by cos 22°:

$$A = d^2 / (2 \cos(22 \text{ deg})) \tag{2}$$

$$HV = F/A = 2F \cos(22 \text{ deg}) / d^2 = 1.8544F / d^2 \tag{3}$$

In Eq. (3), F is in kgf and d is in mm. Diagonals recorded in µm must therefore be divided by 1000 before being used in this equation. The 98.07 N test force is approximately 10 kgf, so the result is written as HV10. This is a macro Vickers test. The designation adds the dwell time only when it falls outside 10-15 s.[1]

Experimental Equipment

- **Safety glasses.** Required eye protection for the lab.

- **Phase II Macro Vickers Hardness Tester, Model 900-398 (Figure 1a).** Applies the test force through a square-based diamond pyramid indenter. After START, it runs the loading, dwell, and

unloading stages on its own, with an indicator light for each stage. The display shows both diagonals, the hardness, the dwell time, and the test force (Figure 1c).

- **Turret with the indenter and objective lenses (Figure 1b).** Swings either the indenter or the 10× objective over the specimen.

- **Measuring eyepiece with two lines.** The left knob moves both lines together and the right knob moves the right line alone. The display reports the diagonal length in increments of 0.01 µm.

- **Test specimens.** 6061-T6 aluminum, 1018 cold-rolled steel, 360 half-hard brass, and 110 copper bars, and a zirconia ceramic disc.

Procedure & Observations

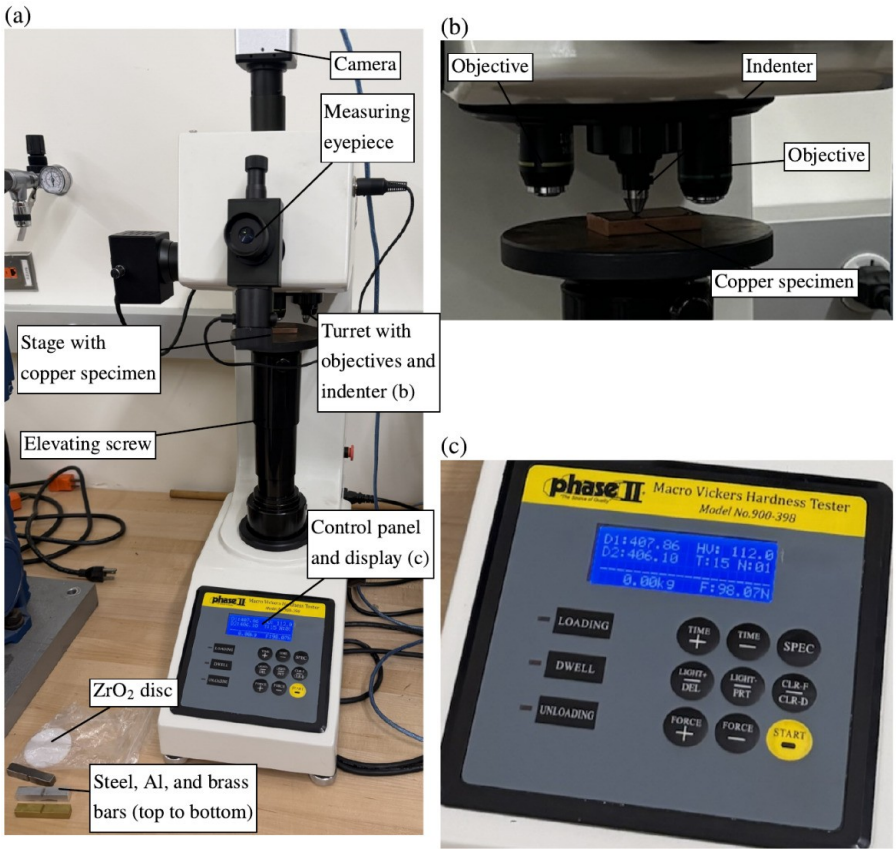


Figure 1. Test setup. (a) Phase II Model 900-398 macro Vickers tester with the specimens on the bench and the copper bar on the stage. (b) Turret with the indenter between the two objective lenses. (c) Control panel after copper indent 1: d_1 = 407.86 µm, d_2 = 406.10 µm, HV 112.0, dwell T = 15 s, and F = 98.07 N.

The steps below are per the lab handout.[2]

1. On the control panel, set the dwell time to 15 s and the test force to 98.07 N (10 kgf).

2. Place the specimen on the stage, swing the 10× objective over it, and focus on the surface at a spot away from earlier indents.

3. Swing the indenter over the specimen and press START. The tester loads, holds the force for the dwell time, unloads, and beeps when the cycle ends.

4. Swing the objective back over the impression. Use the left knob to set both lines on the left corner, then zero the reading (SPEC, then CLR-D).

5. Turn the right knob to move the right line to the right corner and press Measure to record d_1.

6. Rotate the eyepiece 90° without zeroing again, set the lines on the bottom and top corners, and press Measure to record d_2. The display then shows the hardness.

7. Record d_1, d_2, and HV10, and repeat for three indents on each of the five specimens.

Figure 2 shows an impression through the eyepiece with the lines set on its top and bottom corners. Scratches on the specimen surface are visible around the impression. For copper indent 1, the display read d_1 = 407.86 µm, d_2 = 406.10 µm, and HV 112.0 (Figure 1c). Equation (3) gives 111.96 from the same diagonals, which agrees with the tester at its displayed precision. Appendix A lists all thirty recorded diagonals.

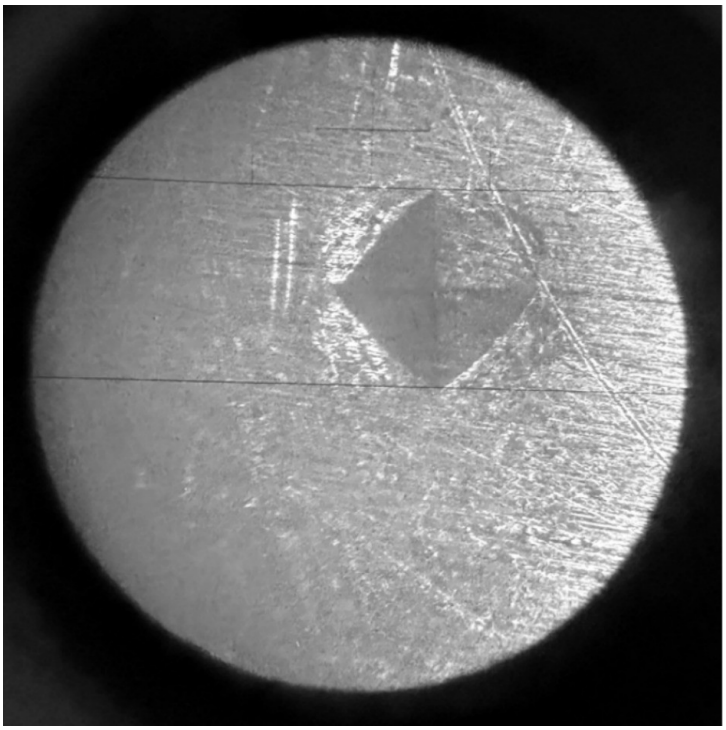


Figure 2. Impression seen through the measuring eyepiece. The two horizontal lines sit on the top and bottom corners of the impression.

Data & Results

The hardness of each indentation and the results for each material are shown in Table 1 and the results are compared in Figure 3. Zirconia was the hardest at 1600 ± 200 HV10, much harder than the steel at 250 ± 10, over 6 times harder. Next came brass at 141 ± 5. Copper and aluminum were almost identical at 108 ± 3 and 107 ± 2 HV10 respectively. The uncertainty ranges of copper and aluminum overlap, so the hardness of the two cannot be clearly distinguished by this test.

Table 1. Vickers hardness (HV10) of each indent, average diagonal, and hardness of each material. The uncertainties come from the average deviation of the six diagonal readings; "s" is the sample standard deviation of the three indent hardness values, used as a cross-check.

Material	HV10 indent 1	HV10 indent 2	HV10 indent 3	Mean diagonal (µm)	HV10 ± delta_HV	s
6061-T6 aluminum	108.6	108.4	104.7	416 ± 3	107 ± 2	2.2
1018 cold-rolled steel	238.5	261.0	251.0	272 ± 6	250 ± 10	11
360 half-hard brass	143.8	137.0	142.0	363 ± 6	141 ± 5	3.6
110 copper	112.0	105.8	105.7	415 ± 5	108 ± 3	3.6
Zirconia (ZrO_2)	1803.3	1293.6	1742.5	108 ± 8	1600 ± 200	280

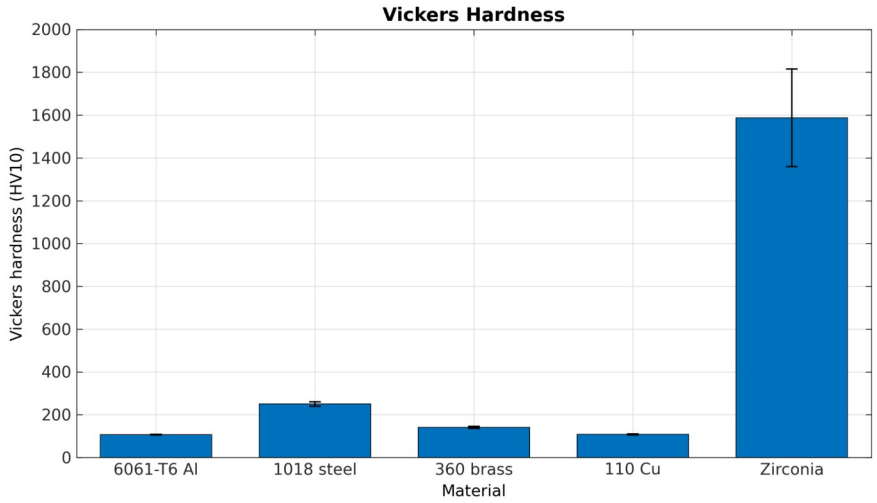


Figure 3. Vickers hardness of each material. Bars show the hardness calculated from the average diagonal with its propagated uncertainty.

Uncertainty Analysis

The display reports each diagonal to 0.01 μm (Figure 1c). This is the display resolution, not a verified accuracy of the measurement. Even using 0.01 μm as a reading increment in Eq. (3), its effect on hardness is at most about 0.02%. The six readings for each material spread much more than that, by about 3-8 μm on average. They combine both diagonals of three separate indents and include variation from setting the lines on the corners.

Following the introductory uncertainty handout, we used the average of the six diagonal readings and their average absolute deviation:[3]

$$d_{avg} = \frac{1}{6}\sum(d_k), \quad \delta_d = \frac{1}{6}\sum(|d_k - d_{avg}|), \quad k = 1 \text{ to } 6 \tag{4}$$

This is an estimate based on the scatter in our readings, rather than a confidence interval or standard error of the mean. There were three independent indentations, with two diagonal measurements per indentation. Following the data table layout, the reported hardness was calculated from the mean diagonal, rather than by averaging the three individual hardness values.

Equation (3) raises the mean diagonal to the power -2. By the power rule, the fractional uncertainty in HV is twice the fractional uncertainty in the diagonal:[3,4]

$$\frac{\delta_{HV}}{HV} = 2\left(\frac{\delta_d}{d_{avg}}\right) \tag{5}$$

Load and calibration uncertainty were not included because those values were not available. The reported uncertainty therefore describes the contribution from diagonal scatter only. As a cross-check, we computed the hardness of each indent from its own mean diagonal and took the sample standard deviation of those three hardness values:[4]

$$s = \sqrt{\frac{1}{n-1}\sum(HV_i - HV_{avg})^2}, \quad i = 1 \text{ to } n, \quad n = 3 \tag{6}$$

Here, the hardness mean in Eq. (6) is the arithmetic mean of the three individual indent hardness values. For every material, s is between 0.77 and 1.32 times the propagated uncertainty

in Table 1. The two estimates are of a similar size, although they describe different measures of scatter. Final values keep one significant figure in the uncertainty, with the hardness rounded to the same place (Rule 5).[3] Appendix B works through the brass calculation.

Discussion

The hardness ranking is consistent with the bonding and strengthening mechanisms discussed in the Introduction. Zirconia was over six times harder than the hardest metal tested, steel. This result is due to the resistance of the ceramic to slip. The composition and cold work could have made the steel hard, but the microstructure was not directly examined in this test. The brass was strengthened by zinc in solid solution in addition to the half-hard temper, and was about 30% harder than copper. Copper and aluminum were within 1 HV10 of each other but differed in their strengthening. The copper finding is consistent with cold work, but the temper was not reported. The hardness of aluminum is controlled by fine precipitates that form during T6 aging.

Specimen preparation may create a hardened surface layer on the bar itself. Nayar notes that sample preparation must not affect surface hardness.[1] The optimal Vickers geometry gives an indentation depth of about d/7, which is about 39 μm for our steel impressions. With this shallow depth, it is worth checking the condition of the surface tested.

The largest fractional uncertainty was for zirconia, at 14.3% before rounding. The impressions were approximately 108 μm across, a quarter of the size of the impressions in aluminum and copper. Thus the same number of micrometers of reading error has about four times the fractional effect. Indents 1 and 3 showed 1803 and 1743 HV10 while indent 2 showed 1294 HV10. The greatest difference between two diagonals was also in zirconia indent 2: 8.0%, compared with 4.2% or less for the other indents. That difference could reflect corner-reading error, alignment, surface condition, or an irregular impression. It does not identify one cause by itself. Ceramic hardness can also depend on test load, and cracking around an impression can affect the diagonal measurement.[1]

There are four further sources of error to consider. First, copper indent 2, steel indent 2, and zirconia indent 3 have identical recorded diagonals to 0.01 µm. Zirconia indent 1 also has the same 103.16 µm reading as indent 3. These values should be checked against the original readings. They could be real, but they could also result from recording one diagonal twice. Matching values alone do not prove a copying error. Second, Nayar recommends spacing the centers of indents by about 6d for light metals such as aluminum and 3d for steel and copper alloys.[1] For our aluminum, 6d is about 2.5 mm. Spacing was not quantified, so interactions between nearby indents cannot be ruled out.

Third, repeated measurements do not necessarily reveal a calibration error that shifts the results in the same direction. Nayar recommends checking the tester against a reference block of similar hardness.[1] A reference-block check was not documented in this report. Fourth, the scratches around the impression in Figure 2 can also make it harder to place the measuring lines on the corners. A smoother surface, a recorded spacing, and a reference-block check would help separate these effects from actual material differences.

Conclusion

We measured the Vickers hardness of five materials at 10 kgf. The results were 107 ± 2 HV10 for 6061-T6 aluminum, 250 ± 10 for 1018 cold-rolled steel, 141 ± 5 for 360 half-hard brass, 108 ± 3 for 110 copper, and 1600 ± 200 for zirconia. Zirconia proved to be the hardest, followed by steel and brass. Copper and aluminum showed similar hardness within the limits of the estimated uncertainty.

Later iterations should record the space between the indents and any cracks in the corners, check a reference block before the specimens and compare each d1 and d2 with the

display before recording them. It would be useful to have more indentations on zirconia to assess the uniformity of its hardness.

References

1. Nayar, A. *Testing of Metals*, Chapter 9, Vickers hardness test. ME 241 pre-lab reading, Fall 2026.

2. "Vickers Hardness Testing Lab." ME 241 Materials Laboratory handout, Department of Mechanical Engineering, San Diego State University, Fall 2026.

3. "Introduction to Uncertainties (prepared for Physics 15 and 17)." Course handout, CCJ, January 8, 1998. Average deviation and Rules 0-5.

4. "Principal Formulas in Part I." Course handout, "Uncertainty propagation." Propagation of uncertainties and statistical definitions.

Appendices

Appendix A. Raw data

Table 2. Diagonals as recorded during the lab (10 kgf, 15 s dwell), with each indent's mean diagonal d_i, signed diagonal difference, and hardness from Eq. (3). The recorded diagonal values are unchanged.

Material	Indent	d_1 (µm)	d_2 (µm)	d_i (µm)	$(d_1 - d_2)/d_i$ (%)	HV10 of indent
6061-T6 aluminum	1	412.64	413.89	413.27	-0.3	108.6
	2	414.14	412.88	413.51	+0.3	108.4
	3	422.43	419.16	420.80	+0.8	104.7
1018 cold-rolled steel	1	277.10	280.62	278.86	-1.3	238.5
	2	266.56	266.56	266.56	0.0	261.0
	3	266.06	277.60	271.83	-4.2	251.0
360 half-hard brass	1	365.19	352.90	359.05	+3.4	143.8
	2	369.97	365.95	367.96	+1.1	137.0
	3	354.66	367.96	361.31	-3.7	142.0
110 copper	1	407.86	406.10	406.98	+0.4	112.0
	2	418.65	418.65	418.65	0.0	105.8
	3	418.66	419.16	418.91	-0.1	105.7
Zirconia (ZrO_2)	1	103.16	99.65	101.41	+3.5	1803.3
	2	114.96	124.50	119.73	-8.0	1293.6
	3	103.16	103.16	103.16	0.0	1742.5

Calculated quantities were evaluated before rounding. HV10 identifies the 10 kgf test force; the numbers in the last column are the hardness of each individual indent.

Appendix B. Sample calculation for brass

The six recorded brass diagonals were 365.19, 352.90, 369.97, 365.95, 354.66, and 367.96 µm. The mean diagonal is:

$$d_{avg} = (365.19 + 352.90 + 369.97 + 365.95 + 354.66 + 367.96)/6$$

$$d_{avg} = 362.7717 \text{ um} = 0.3627717 \text{ mm}$$

Using Eq. (4), the average absolute deviation is:

$$\Delta d = (1/6) \sum(|d_k - 362.7717|) = 5.9944 \text{ um}$$

Substituting the mean diagonal and the 10 kgf force into Eq. (3):

$$HV = 2(10) \cos(22^\circ) / (0.3627717)^2 = 140.91$$

Then Eq. (5) gives:

$$\Delta HV = 140.91(2)(5.9944/362.7717) = 4.66$$

After rounding the uncertainty to one significant figure and the hardness to the same place, the brass result is:

$$HV10 = 141 \pm 5$$

As a check, brass indent 1 has a mean diagonal of (365.19 + 352.90)/2 = 359.045 um, giving 143.85 HV10 before rounding. The three indent hardness values are 143.8461, 136.9603, and 142.0483 HV10. Their sample standard deviation is 3.57 HV10, reported as 3.6 in Table 1.